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## Your grade: 100%

Your latest: **100%** • Your highest: **100%** • To pass you need at least 80%. We keep your highest score.

# Full Course Assessment

**Review Learning Objectives**

**Assignment details**

**Due**
Jan 10, 11:59 PM IST

**Attempts**
Unlimited

**Retry**

**Submitted**
Dec 28, 3:04 AM IST

**Your grade**
**100%**

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**1.** You want to create a nested list in Python where each sublist contains a string and a number. Which code snippet correctly achieves this? **1 / 1 point**

- (•) nested_list = [ ['apple', 1] , ['banana', 2] ]
- ( ) nested_list = 'apple', 1, 'banana', 2
- ( ) nested_list = ['apple', 1], ['banana', 2]
- ( ) nested_list = ['apple': 1, 'banana': 2]

**Correct**
Correct! This syntax creates a list containing sublists with a string and a number.

**2.** What is the main difference between a list and a set in Python? **1 / 1 point**

- ( ) Both lists and sets maintain order, but only lists allow duplicates.
- (•) A list maintains order and allows duplicates, while a set does not maintain order and does not allow duplicates.
- ( ) Both lists and sets do not maintain order, but only sets allow duplicates.
- ( ) A set maintains order and allows duplicates, while a list does not maintain order and does not allow duplicates.

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**Correct**
Correct! Lists maintain order and allow duplicates, whereas sets do not maintain order and do not allow duplicates.

**3.** Which of the following are considered immutable in Python? **1 / 1 point**

- [x] String

**Correct**
Correct! Strings are immutable; once created, their characters cannot be changed.

- [ ] List
- [ ] Set
- [ ] Dictionary
- [x] Tuple

**Correct**
Correct! Tuples are immutable, meaning their elements cannot be changed after creation.

**4.** How can you create a dictionary with different data types as keys in Python? **1 / 1 point**

- ( ) By using square brackets and separating key-value pairs with commas
- (•) By using curly braces and separating key-value pairs with colons
- ( ) By using angle brackets and separating key-value pairs with dashes
- ( ) By using parentheses and separating key-value pairs with semicolons

**Correct**
Correct! Dictionaries in Python are created using curly braces with key-value pairs separated by colons.

**5.** Which method would you use to split a string into a list of substrings in Python? **1 / 1 point**

- (•) split()
- ( ) cut()
- ( ) divide()
- ( ) slice()

**Correct**
Correct! The `split()` method splits a string into a list of substrings based on a specified delimiter.

**6.** Which of the following statements about for loops in Python are true? **1 / 1 point**

- [x] For loops can be used to iterate over items in a list.

**Correct**
Correct! For loops can iterate over items in a list.

- [ ] For loops can only be used with lists.
- [ ] For loops always require an else statement.
- [x] For loops can be used to perform operations on tuples.

**Correct**
Correct! For loops can also be used to iterate over and perform operations on tuples.

- [x] For loops can be used to iterate over dictionary keys and values.

**Correct**
Correct! For loops can iterate over both keys and values in a dictionary.

**7.** What will be the output of the following Python code snippet? **1 / 1 point**

```
1
2  my_list = [0, 1, 2, 3, 4]
3  filtered_list = [x for x in my_list if x % 2 == 0]
4  print(filtered_list)
5
```

- ( ) [ ]
- ( ) [1, 3]
- (•) [0, 2, 4]
- ( ) [0, 1, 2, 3, 4]

**Correct**
Correct! The list comprehension filters out only the even numbers.

**8.** Which of the following are valid ways to use the `break` and `continue` keywords in Python loops? **1 / 1 point**

- [x] Using `break` to exit a loop when a condition is met.

**Correct**
Correct! The `break` keyword is used to exit a loop when a certain condition is met.

- [ ] Using `break` to skip to the next iteration of the loop.
- [ ] Using `continue` to exit the loop when a condition is met.
- [x] Using `continue` to skip the current iteration and continue with the next iteration.

**Correct**
Correct! The `continue` keyword skips the current iteration and moves to the next iteration in the loop.

**9.** Which of the following statements about using `if` statements with lists and dictionaries in Python are true? **1 / 1 point**

- [x] You can use `if` statements to check if a specific value is present in a list.

**Correct**
Correct! `if` statements can check for the existence of values in lists.

- [x] You can use `if` statements to check if a specific key exists in a dictionary.

**Correct**
Correct! `if` statements can indeed check for the existence of keys in dictionaries.

- [ ] You can use `if` statements to compare two dictionaries directly.
- [ ] You can use `if` statements to check if a list is empty by comparing it to `None`.

**10.** Which of the following methods can you use to combine multiple lists into a single list in Python? **1 / 1 point**

- ( ) concatenate()
- (•) zip()
- ( ) append()
- ( ) merge()

**Correct**
Correct! The `zip()` method can be used to combine multiple lists into one.

**11.** What is the purpose of the 'self' keyword in Python classes? **1 / 1 point**

- ( ) To import modules
- ( ) To define a global variable
- (•) To refer to the instance of the class
- ( ) To call a superclass method

**Correct**
Correct! The 'self' keyword is used to refer to the instance of the class within its methods.

**12.** Which of the following best describes how to use the 'global' keyword in Python? **1 / 1 point**

- ( ) To restrict a variable to a function scope only.
- ( ) To create a new global variable inside a function.
- (•) To declare a variable inside a function as global.
- ( ) To define a variable that can only be read inside a function.

**Correct**
Correct! The 'global' keyword is used to declare a variable inside a function as global so that it can be modified outside the function scope.

**13.** What is the purpose of the `name` variable in a Python module? **1 / 1 point**

- (•) To specify the module's name when it is executed directly.
- ( ) To define the global namespace of a module.
- ( ) To specify the return type of the module.
- ( ) To store the module's metadata information.

**Correct**
That's correct! The `name` variable is used to check if a module is executed directly or imported.

**14.** Which of the following statements about the LEGB rule in Python are correct? **1 / 1 point**

- [x] Built-in scope is searched after Global scope

**Correct**
Correct! Built-in scope is searched last after Local, Enclosing, and Global scopes.

- [ ] Enclosing scope is the scope of the module
- [ ] LEGB determines the order of operations in mathematical expressions
- [x] Local scope is searched first according to the LEGB rule

**Correct**
Correct! Local scope is indeed searched first according to the LEGB rule.

- [x] LEGB stands for Local, Enclosing, Global, Built-in

**Correct**
Correct! LEGB stands for Local, Enclosing, Global, Built-in, which are the scopes Python searches for variables.

**15.** Which of the following are correct ways to handle errors in Python? **1 / 1 point**

- [ ] Breaking the code with an intentional error.
- [x] Using try-except blocks.

**Correct**
Correct! The try-except block is the standard way to handle errors in Python.

- [x] Using assertions.

**Correct**
Correct! Assertions can be used to confirm that a condition holds true at a certain point in the program.

- [ ] Ignoring the error and letting the program crash.

**16.** Which of the following are valid ways to import a module in Python? **1 / 1 point**

- [x] import numpy as np

**Correct**
Correct! 'import numpy as np' imports the numpy module with an alias 'np'.

- [ ] include numpy
- [x] from numpy import *

**Correct**
Correct! 'from numpy import *' will import all functions and classes from the numpy module.

- [ ] using numpy
- [x] import numpy

**Correct**
Correct! 'import numpy' is a standard way to import the numpy module.

**17.** What does the 'global' keyword do in Python? **1 / 1 point**

- ( ) It imports global variables from other modules.
- ( ) It declares a variable as global throughout the entire project.
- (•) It allows modification of a global variable within a function.
- ( ) It creates a new global variable.

**Correct**
Correct! The 'global' keyword allows you to modify a variable outside of the current scope.

**18.** Which of the following are purposes of the special variable `name` in Python? **1 / 1 point**

- [x] To determine if the module is being run directly or imported.

**Correct**
Correct! The special variable `name` is used to check if the module is being run directly or imported.

- [x] To execute code only if the file is run as a main program.

**Correct**
Correct! Code within the if `name` == '`main`' block is executed only if the file is run as a main program.

- [ ] To specify the name of a function.
- [ ] To define the main function of a class.

**19.** Which of the following best describes the purpose of the `init` method in Python? **1 / 1 point**

- ( ) To print the attributes of an object
- ( ) To call a superclass initializer
- (•) To initialize an object's attributes
- ( ) To delete an object's attributes

**Correct**
Correct! The `init` method is used to initialize the attributes of an object when it's created.

**20.** What is the purpose of the 'filter' function in Python? **1 / 1 point**

- ( ) To transform each element of a list through a function.
- ( ) To iterate through elements of a list.
- ( ) To sort elements in a list.
- (•) To select elements from a list that meet a specific condition.

**Correct**
Correct! The 'filter' function is used to filter elements from a list based on a specific condition.